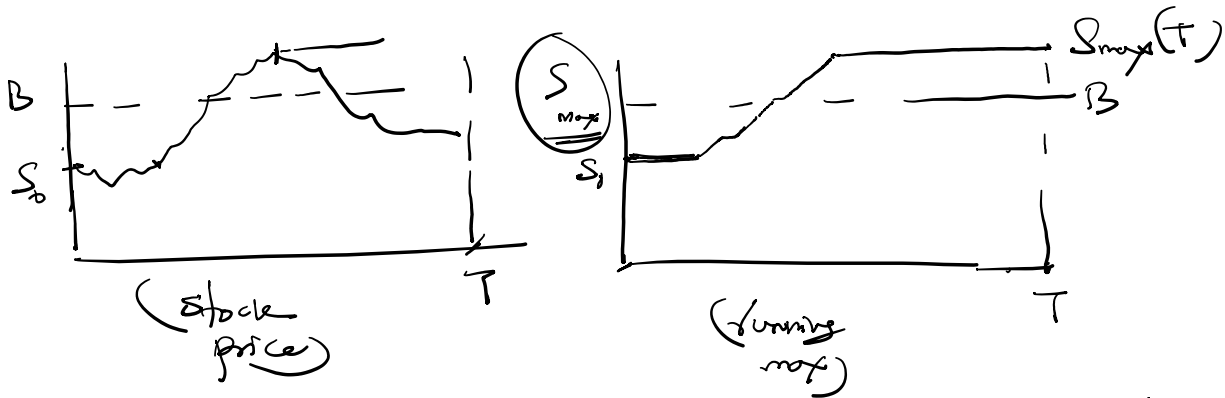
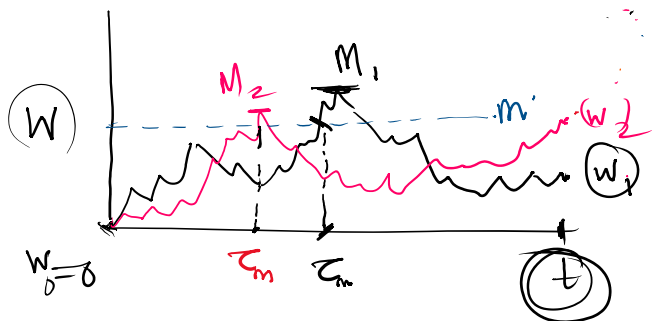




$$1_{S_{max}(T) \geq B} \rightarrow 1, \quad S_{max}(T) \geq B$$

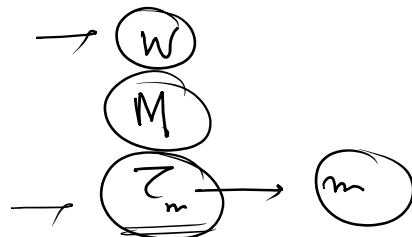
$$1_{S_{max}(T) < B} \rightarrow 0, \quad S_{max}(T) < B$$



$$\underline{M(t)} = \max_{0 \leq s \leq t} W_s$$

$$\underline{W, M}$$

$$\begin{aligned} &\rightarrow W(t) = W_1 \\ &\rightarrow M(t) = M_1 \\ &W(t) = W_2 \\ &M(t) = M_2 \end{aligned}$$

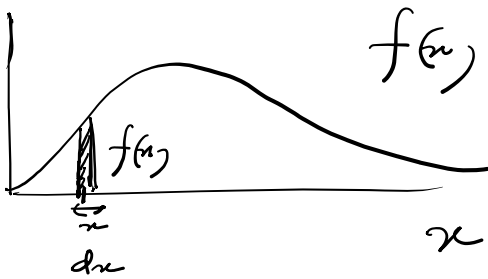


First passage time $\underline{\tau_m}$

$$\underline{P(\tau_m \leq t)} = \underline{P(M(t) \geq m)}$$

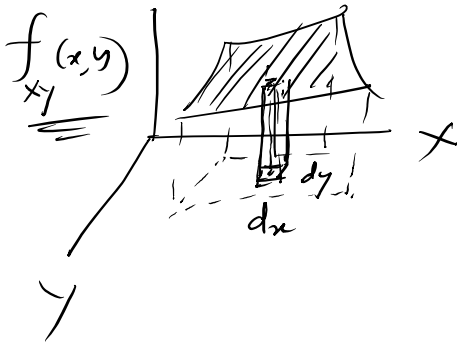
$\rightarrow$ m has been reached before t

(barrier has been crossed before t)



$$E(X) = \int x \cdot dP = \int \underline{x} f(x) dx$$

$$dP = \underline{f(x) dx}$$



$$\underline{dP} = \underline{f(x, y) dx dy}$$

$$\underline{P(X \leq x, Y \leq y)} = \int_{-cb}^y \int_{-cb}^x \underline{f(x, y) dx dy}$$

$$E(g) = \iint g(x, y) \cdot f(x, y) dx dy$$

